



Unemployment Trends During COVID-19 in India

Exploratory Analysis and Explainable Machine Learning

This dashboard explores unemployment trends across India before and during COVID-19 using statistical analysis, visualization, XGBoost and SHAP explainability.

Average Unemployment

11.79%

Maximum Unemployment

76.74%

Pre-COVID Mean

9.61%

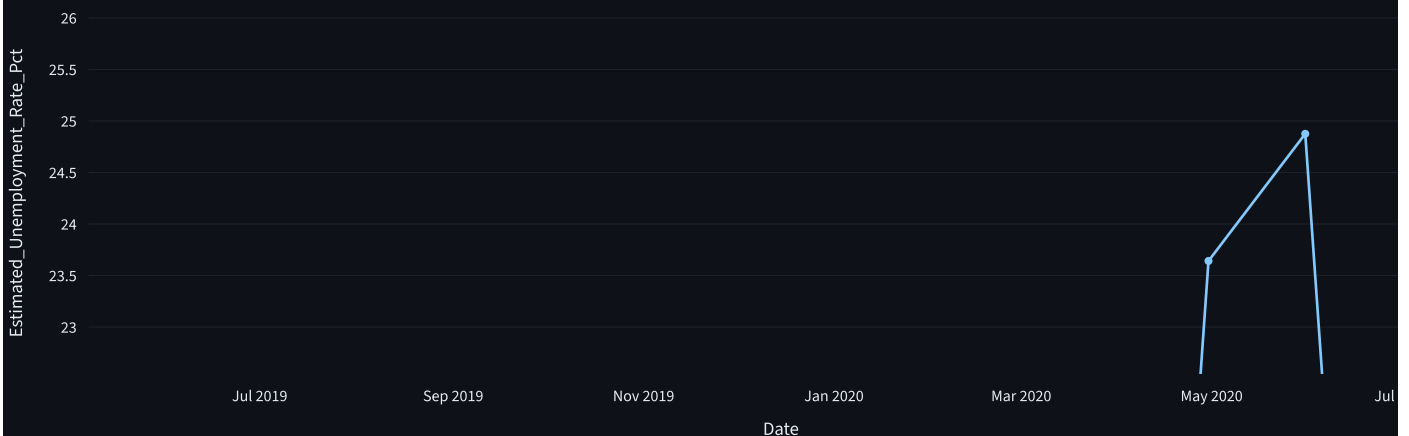
COVID Increase

110.0%



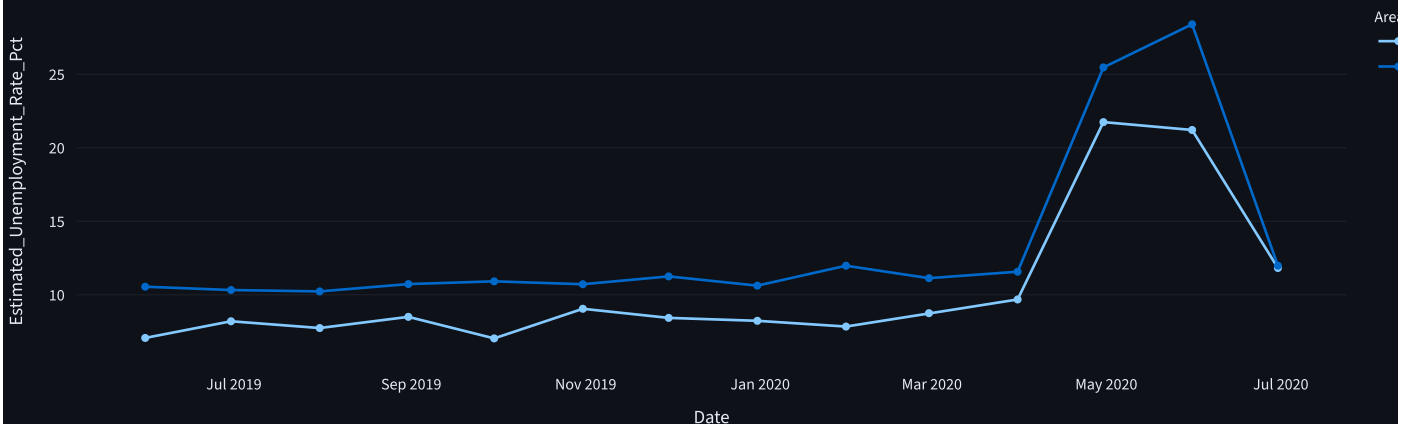
National Unemployment Trend

National Average Unemployment Rate



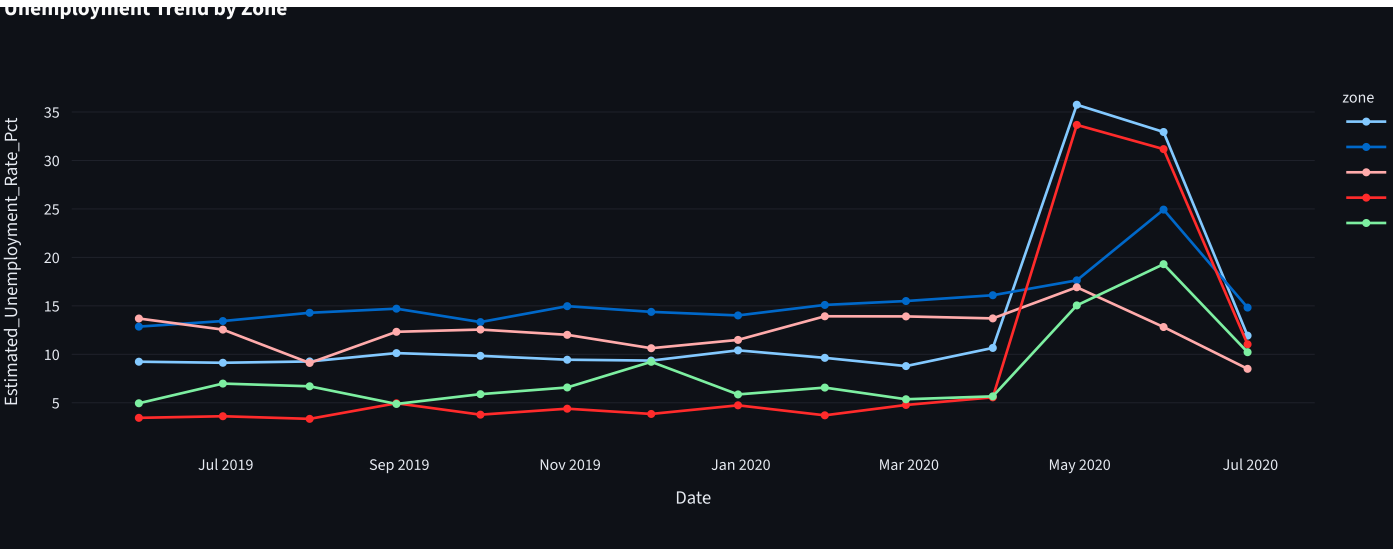
Urban vs Rural Comparison

Urban vs Rural Unemployment Trend

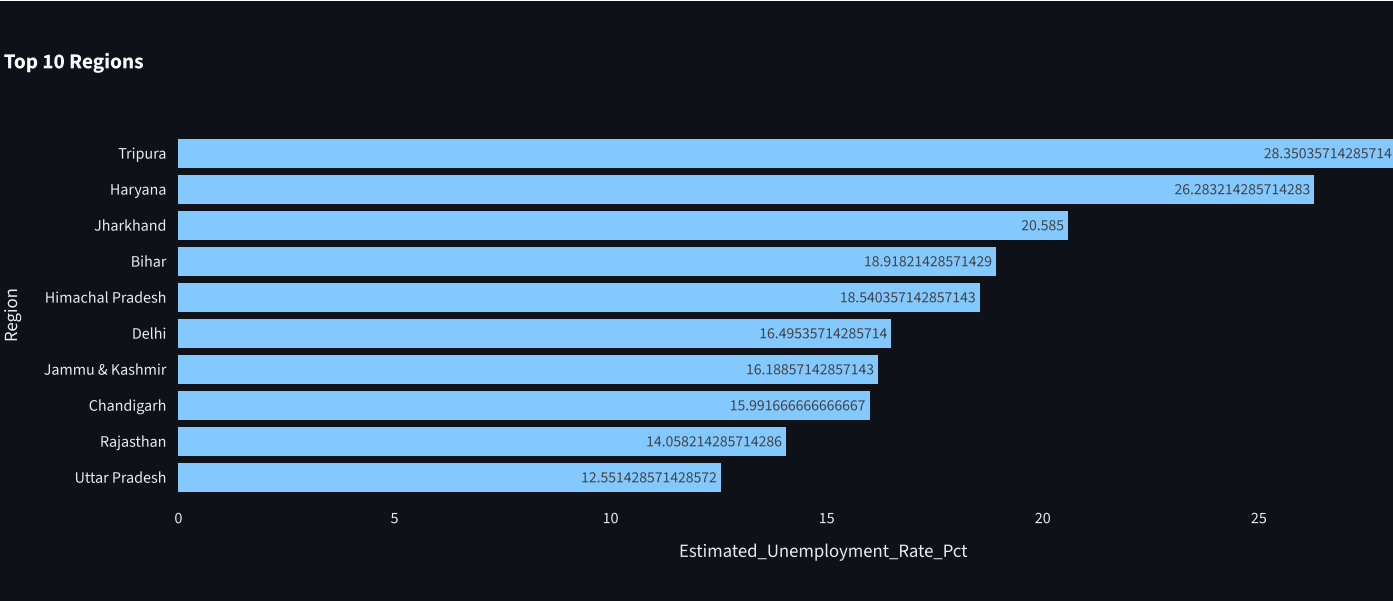


Zone-Level Unemployment Trend

Unemployment Trend by Zone



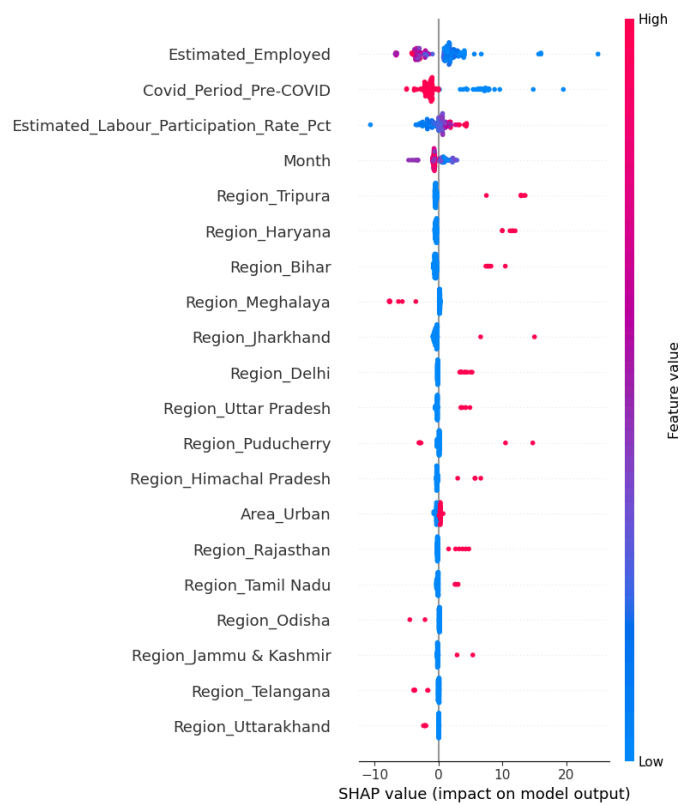
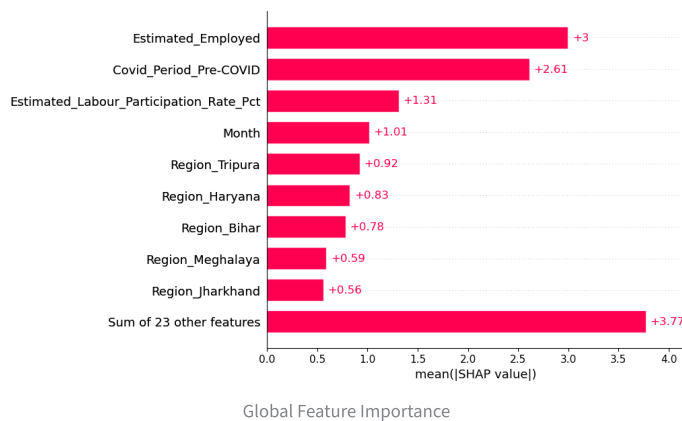
🏆 Top 10 Regions by Average Unemployment



🔗 XGBoost Model Performance

	Metric	Value
0	MAE	
1	RMSE	
2	R ²	

🔍 Explainable AI (SHAP)



Key Findings

Main Insights

- COVID-19 increased average unemployment by approximately **110%**.
- The increase was statistically significant ($p < 0.001$).
- Urban areas experienced consistently higher unemployment than rural areas.
- Northern and Eastern regions showed the largest unemployment spikes.
- Tripura, Haryana, Jharkhand and Bihar were among the most affected regions.
- XGBoost explained approximately **72%** of unemployment variation.
- SHAP analysis identified employment level, COVID period and regional effects as the strongest drivers of unemployment.